TIOTROPIUM

Once daily

Capsules for inhaler only

CHEMICAL NAME (1R,2S,4R,5S,7s)-7-[(hydroxy)(di-2-thienyl)acetoxy]-9,9-dimethyl-3-oxa-9-azoniatricyclo[3.3.1.0^{2},^{4}]nonane bromide (monohydrate)

CLASS Long-acting muscarinic antagonist (LAMA)

TRADE NAMES Spiriva HandiHaler · Spiriva Respimat · Spiolto (with olodaterol)

THERAPEUTIC USES / INDICATIONS

- COPD maintenance — reduces exacerbations and improves lung function.
- Asthma add-on (Respimat) in adults/children ≥6 yr uncontrolled on ICS/LABA.

PHARMACODYNAMICS

- Long-lasting blockade of **M3 receptors** in airways → sustained bronchodilation.
- Reduces hyperinflation and exacerbations.

ADMINISTRATION

- **HandiHaler:** place capsule in device, pierce once, inhale slowly and deeply twice — capsules must not be swallowed.
- **Respimat:** soft-mist inhaler, 2 puffs once daily.
- Same time each day; not for acute relief.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth.
- Constipation, cough.
- Sinusitis, pharyngitis.

ADVERSE EFFECTS (SERIOUS)

- Urinary retention.
- Angle-closure glaucoma.
- Tachyarrhythmias (caution).
- Paradoxical bronchospasm.

DOSAGES

- **HandiHaler:** 18 mcg once daily (1 capsule).
- **Respimat:** 5 mcg (2 puffs of 2.5 mcg) once daily.
- **Asthma (Respimat):** 2.5–5 mcg once daily.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to atropine derivatives or lactose (HandiHaler, milk protein).
- **Caution:** narrow-angle glaucoma, prostatic hyperplasia, recent MI, arrhythmias, CrCl ≤50.

KEY INTERACTIONS

- **Ipratropium and other anticholinergics:** avoid combination.

PHARMACOKINETICS

- **Onset:** 30 min; peak 3 h. **Duration:** >24 h.
- Low systemic bioavailability; slow dissociation from M3 receptors.
- Excreted in urine. **t½:** 5–6 days (terminal).

NURSING MANAGEMENT

- Teach and check inhaler technique.
- Stop regular ipratropium when tiotropium started.
- Assess dyspnoea, exacerbation frequency, spirometry.
- Ask about urinary symptoms, eye pain.

PATIENT EDUCATION

- Use once daily at the same time — not for sudden breathlessness.
- Never swallow the capsules.
- Rinse mouth or sip water for dryness.
- Report eye pain, blurred vision or difficulty passing urine.

KEY POINTS

- LAMA; once daily COPD maintenance.
- Not a reliever.
- Capsules for inhalation only.
- Do not combine with ipratropium.

SOURCES CHECKED **[455]** Spiriva Respimat, inhaled, (inhalation) (US label) **[456]** Tiotropium Bromide, oral, inhaled, (inhalation) (US label)

Full references p. 432–457